Ex. : 3
Date: 23/7/26

understanding CISCO PACKET TRACKER

Aim:-

To understand the cisco packet Tracker environment and begin design a simple computer network using its interface and networking components.

steps:

1. open the cisco packet tracker application
2. observe the VI including mane Bar, main tool bar, workspace and other.
3. From the network component base, select the required network device such as pc's HOB and Switch
4. Drag and drop the devices into the workspace
5. correct the devices using copper straight through cables
6. configure IP address for the PC using the Desktop -> IP configuration option.
7. Save the network topology.

Result:

The cisco packet tracker environment was Successfully explored and a simple network was designed. the interface components, device connections, IP configuration and packet Tracker was Successfully understand using both realtime and Stimulation modules.

Exc : 3.1

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HUB

Aim:

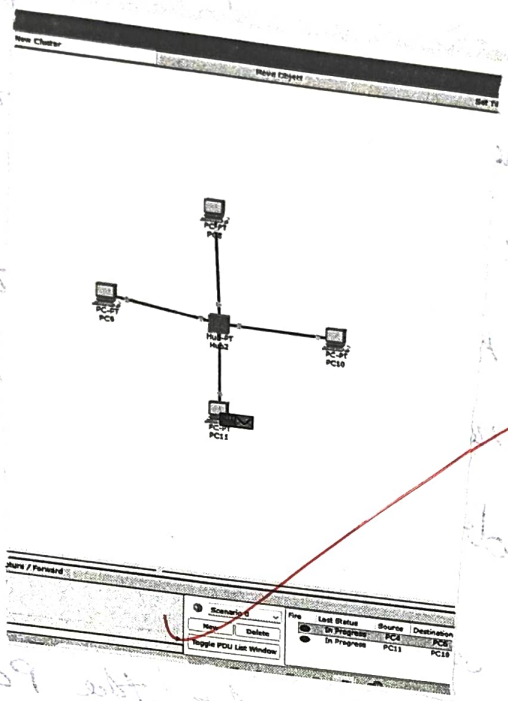
To study the behaviour of a HUB by observing how its forwards data packets using packet Tracer.

Steps:

* open cisco packet Tracer.
* Drag and place 1 HUB and 4 Generic PC's into the workspace.
* connect the all PC's to the HUB using copper straight through cables.
* configure PC addresses for all PC's in the same network.
* Select Add simple PDU from the comon Tool Bar.
* Send a packet from one PC (source) to another PC (destination)
* Switch to Simulation mode and observe how the HUB forwards the packets.

Result:-

The HUB broadcasts the received packet to all connected devices. Every PC Receives the packet but only distination PC accepts it while the others discard it.


New Cluster
New Object
Set Ti
PC0
PC3
Hub-PT
Hub0
PC10
PC11
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
In Progress
PC4
PC11
PC10

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Switch

Aim:-

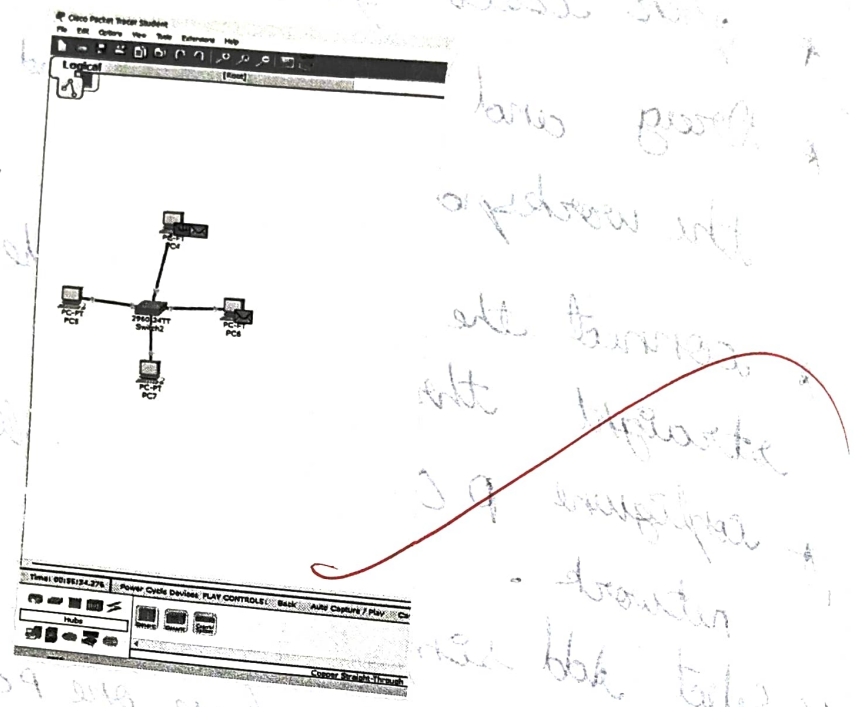
To study the behaviour of a switch by observing how it forwards data packets using cisco packet tracker.

Steps:

* open cisco packet Tracker
* Drag and place 1 Switch and 4 Generic PC onto workspace.
* Configure IP address for all PC's
* Connect all PC's to the Switch using copper straight through cables.
* Select add simple PDU from common tool bar.
* send a packet from one PC (source) to another PC (destination)
* Switch to simulation mode and observe how the Switch forwards the packet.

Result:-

The Switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.


Cisco Packet Tracer Student
Logical
PC-PT PC1
PC-PT PC2
2950-24TT Switch1
PC-PT PC3
PC-PT PC4
Power Cycle Devices
FAST CONTROLE
Basic
Auto Capture / Play
hubs
Copper Straight-Through

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HUB and Switch (comparison)

Aim:

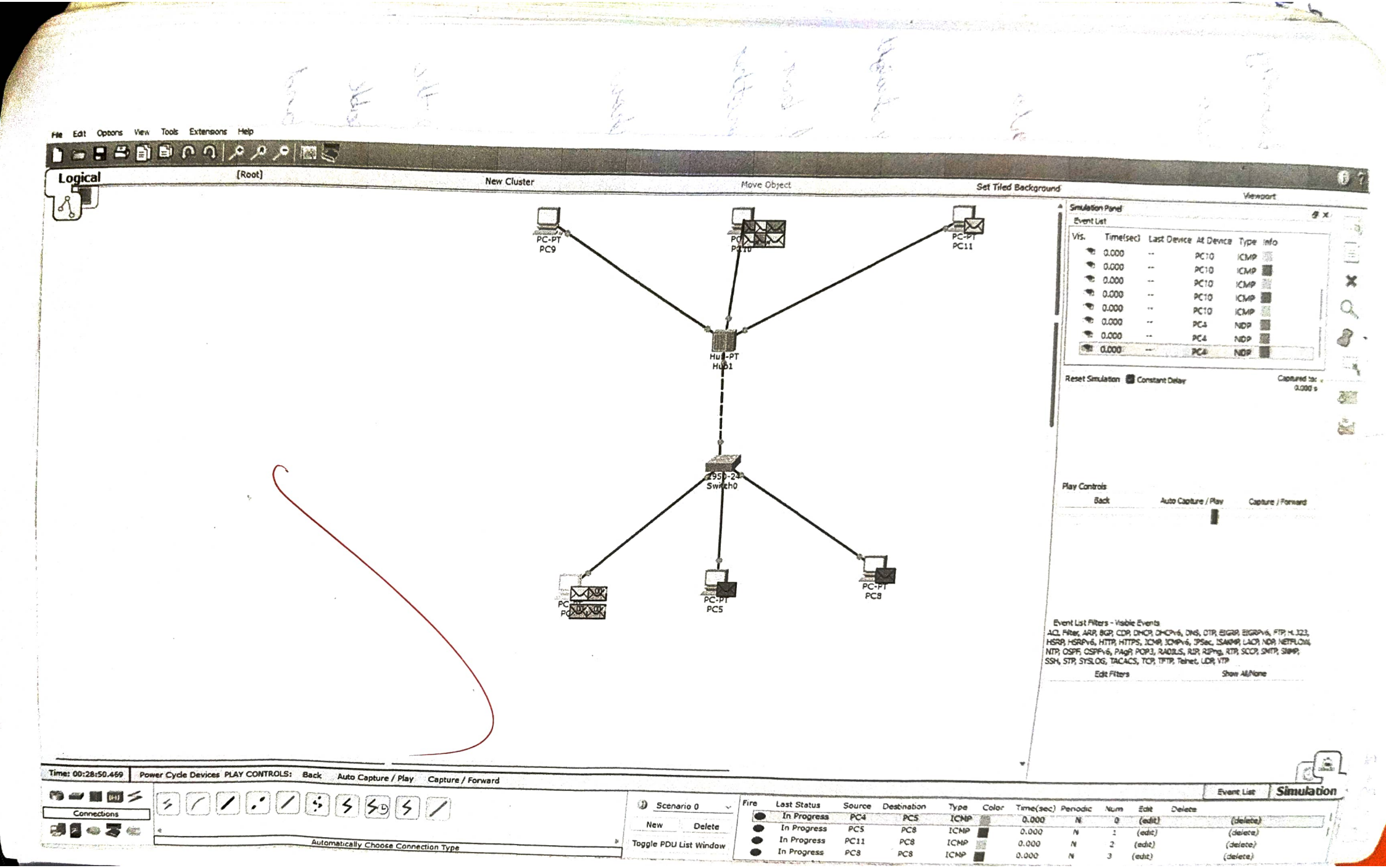
To compare the packet forwarding behaviour of a HUB and a Switch using cisco packet tracer.

steps:

1. create one network using a HUB and another using a Switch, each connects to HPC's
2. configure IP address for all PC's in both network.
3. send a simple PDU between two PC's in each network.
4. observe packet transmission in simulation mode
5. compare how the HUB and switch forward packets.

Result:

The HUB broadcasts packets to all connected devices, whereas the Switch forwards packets only to the intended destination, therefore switch provides better performance, reduced network traffic and improve efficiency compared to the HUB.


File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
PC-PT PC9
PC-PT PC10
PC-PT PC11
Hub-PT Hub1
2950-24 Switch0
PC-PT PC5
PC-PT PC8
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type Info
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC4 NDP
0.000 -- PC4 NDP
0.000 -- PC4 NDP
Reset Simulation
Constant Delay
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters - Visible Events
Edit Filters
Show All/None
Time: 00:28:50.469
Power Cycle Devices
PLAY CONTROLS:
Back
Auto Capture / Play
Capture / Forward
Connections
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Event List
Simulation
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
In Progress PC4 PC5 ICMP 0.000 N 0 (edit) (delete)
In Progress PC5 PC8 ICMP 0.000 N 1 (edit) (delete)
In Progress PC11 PC8 ICMP 0.000 N 2 (edit) (delete)
In Progress PC8 PC8 ICMP 0.000 N 3 (edit) (delete)